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TreeDiff: AST-Guided Code Generation with Diffusion-based LLMs (DLLMs)

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🎯 **TreeDiff** = first AST-aware masking for diffusion code generation → **+13.3% on HumanEval+**, narrows gap with autoregressive models.

Motivation

Diffusion-based LLMs (DLLMs) struggle with **syntactic precision** in code generation. Standard random token masking:

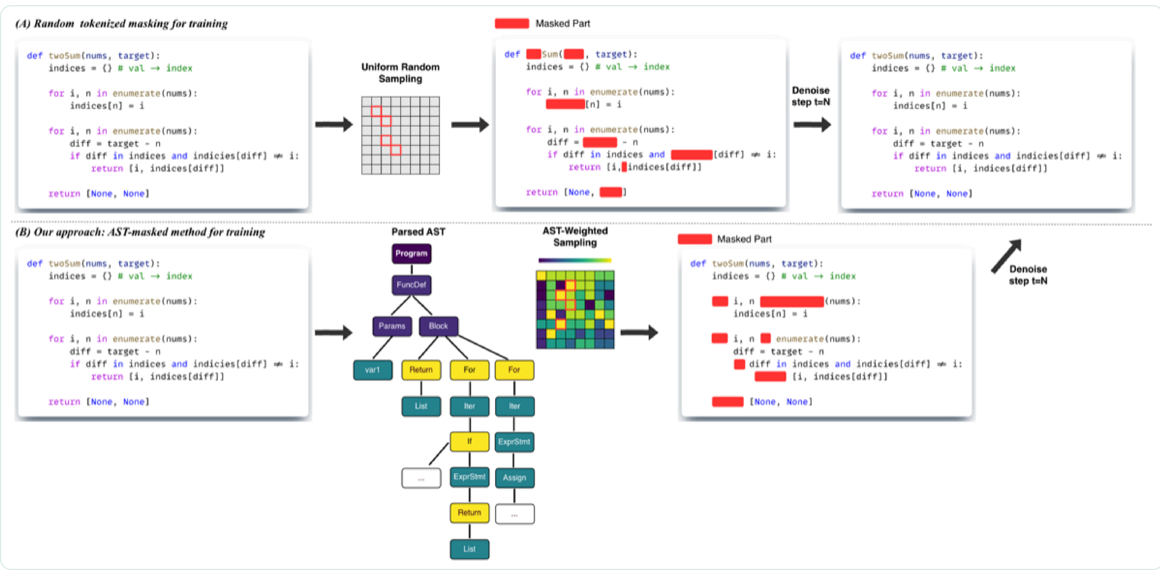
- Ignores syntactic boundaries during denoising
- Fails to capture long-range hierarchical dependencies
- Treats all tokens equally — but they're not!

🔑 **CORE INSIGHT**

"For code, noise should not be purely stochastic — structural priors enable DLLMs to learn program-level dependencies."

Method: AST-Guided Masking

TreeDiff parses code into an **Abstract Syntax Tree (AST)** and uses it to guide which tokens to mask — targeting structurally critical nodes.



TreeDiff augments masked diffusion with **AST-aware policies**: the syntax tree guides which tokens to mask and how to weight the denoising loss.

Region-Specific Masking

Each diff region gets a tailored masking policy:

Prompt	Reasoning	Code
intact	random	AST-weighted

AST Node Weighting

Loss weights scale with AST node depth: **deeper nodes receive higher weight** than surface tokens.

```
w(node) = 1 + α · depth(node)
```

Category	Examples	Weight	Purpose
Skeletal	Imports, Constants	0.15	Preserve skeleton
Data Flow	Assigns, Calls	0.42	Moderate priority
Conditionals	if, elif, else	0.58	Core logic
Control Flow	for, while, return	**0.60**	⚡ Highest priority

Main Results (Pass@1 %)

+13.3%
Relative Gain
HumanEval+ (T=512)

42.1
HumanEval Pass@1
(vs 39.6 baseline)

Model	T = 256				T = 512			
	HE	HE+	MBPP	MBPP+	HE	HE+	MBPP	MBPP+
Autoregressive Models†								
DeepSee... Coder-33B	50.6	44.5	80.4	70.1	50.6	44.5	80.4	70.1
CodeQw... 7B	51.8	45.7	73.5	60.8	51.8	45.7	73.5	60.8
CodeLla... 13B	42.7	38.4	63.5	52.6	42.7	38.4	63.5	52.6
Diffusion-based LLMs (LLaDA Backbone)								
LLaDA-Instruct	39.6	34.8	51.9	43.1	36.6	31.7	39.4	31.7
+ Random Masking	39.6	35.4	52.9	43.9	38.4	32.3	41.5	32.8
Ours								
+ TreeDiff	**42.1**	**37.2**	**52.9**	**44.2**	**40.2**	**36.6**	**42.3**	**33.3**

† From EvalPlus Leaderboard. HE = HumanEval, HE+ = HumanEval+.

Ablation: Masking Granularity

HumanEval

Strategy	T=256	T=512
Random Masking	39.6 / 35.4	38.4 / 32.3
AST (Span)	40.9 / 36.6	40.2 / 36.6
AST (Node)	**42.1 / 37.2**	**40.2 / 36.6**

MBPP

Strategy	T=256	T=512
Random Masking	52.9 / 43.9	41.5 / 32.8
AST (Span)	51.6 / 42.9	39.7 / 31.5
AST (Node)	**52.9 / 44.9**	**42.3 / 33.3**

💡 **KEY INSIGHT**

*Random masking outperforms span masking on code diffs. Spans often split syntactic units (e.g., an if-condition), while random masking preserves more complete AST nodes. TreeDiff's **token-level** AST masking surgically targets high-influence nodes while preserving context.*

Robustness Analysis

Method	T=512	T=1024
LLaDA + span mask	35.6	34.2
TreeDiff	**42.1**	**41.8**

TreeDiff maintains stable performance (-0.1%) as T increases 256→512→1024, while baselines degrade by ~2.1% due to fragmented syntax trees.

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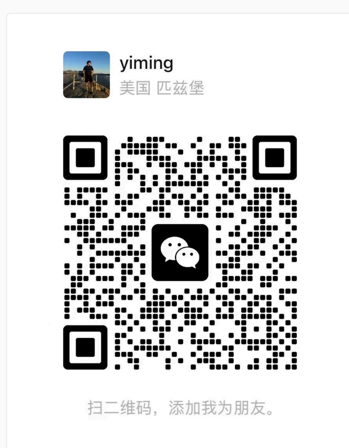

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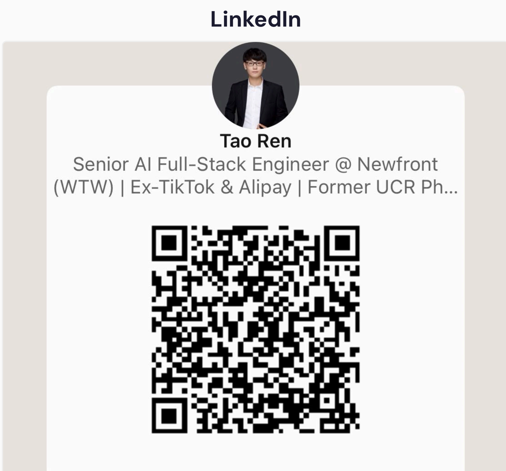

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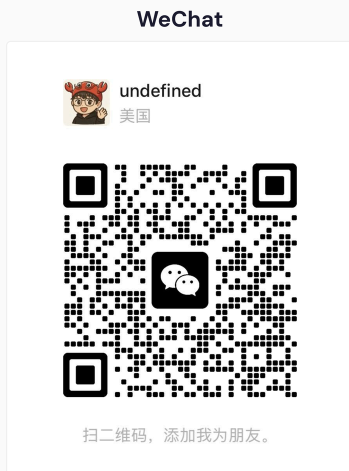
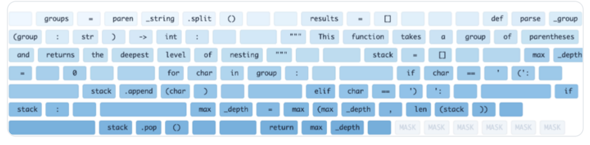


We welcome collaboration and discussion. Feel free to reach out!

Generation Trajectory Analysis

Task: Max nesting depth of parentheses (HumanEval/6), Step 110/256.

Vanilla MD (Random Mask): Uses shallow counter heuristic, fails on nested structures. Fragmented generation.



TreeDiff: Establishes stack-based control flow early. Correctly captures recursive nesting.

Edit distance across denoising steps. Vanilla MD oscillates (54 → 52 → 41 → 46 → 32). TreeDiff converges monotonically with fewer steps.

Only 50 denoising steps (vs 256 decode steps) needed

Generated Code Comparison

HumanEval/6 — Nested parentheses depth:

Vanilla MD (Random Mask)

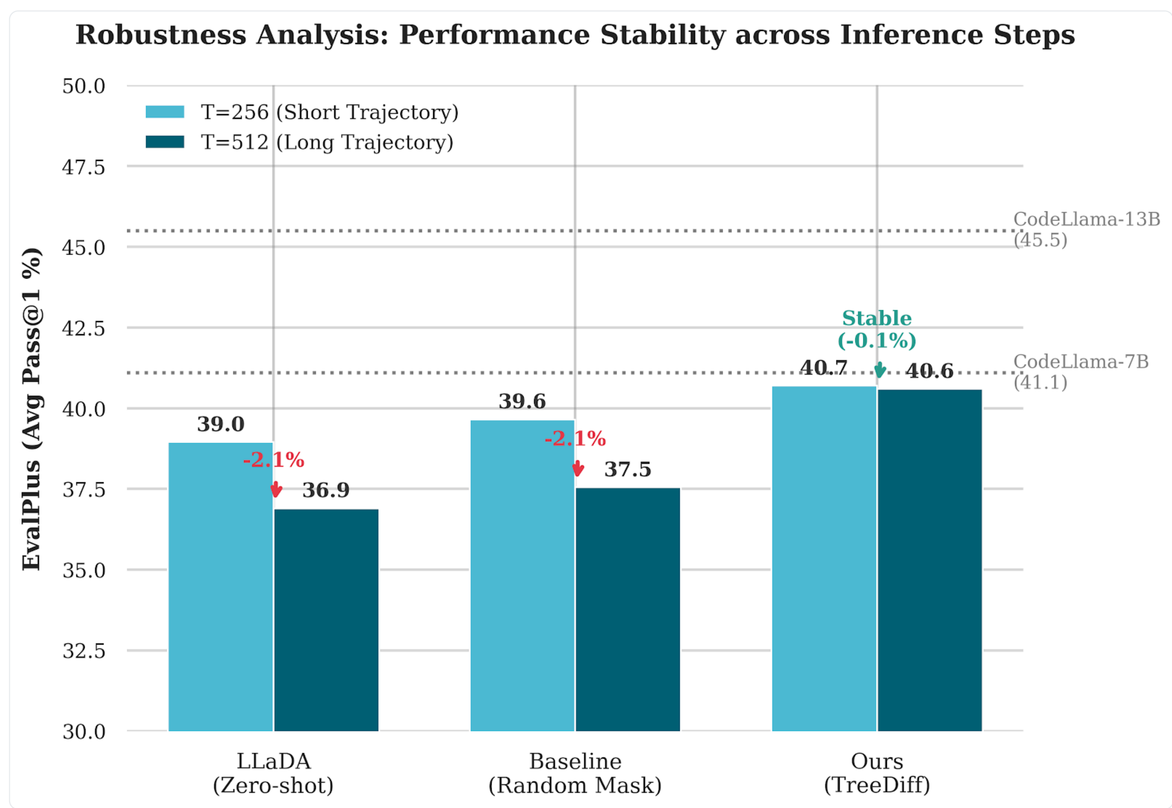
```
count = 0
for char in group:
    if char == '(':
        count += 1
    # Fails to track
    # nested peak depth
```

TreeDiff (AST-guided)

```
for char in group:
    if char == '(':
        stack.append(char)
        max_depth = max(
            max_depth, len(stack))
    elif char == ')':
        if stack: stack.pop()
```

HumanEval/6: Maximum nesting depth of parentheses

EvalPlus Robustness



TreeDiff maintains stable EvalPlus Pass@1 across inference steps (T=256→512), while baselines degrade by ~2.1%.

Key Contributions & Conclusion

- **First AST-aware masking** for DLLMs in code generation
- Novel AST-aware masking and loss weighting that amplifies structural signal
- Achieve **+13.3%** relative HumanEval+ and **42.1** Pass@1 with only 50 denoising steps — lower FLOPs than 256-step autoregressive decoding
- Practical recipe for adapting pretrained masked diffusion LMs to structured seq2seq tasks without catastrophic forgetting

TreeDiff addresses a fundamental limitation: **random masking ignores code structure**. By aligning corruption with AST hierarchy, the model learns programming language compositionality, reconstructing programs that respect grammatical boundaries.